PR-AIE

Day-3 - Python - feb 26

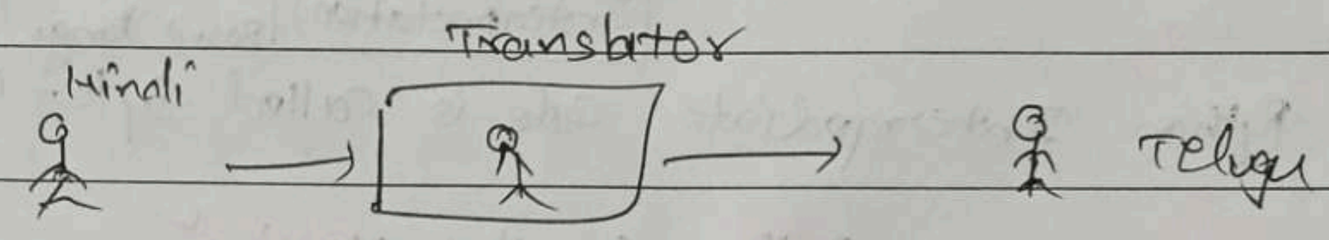
→ Python is a Interpreted Language.

→ 1. compiler & Compilation [C, C++]

2. Interpreter & Interpretation [Python]

3. Performance.

→ Compiler & Interpreter :

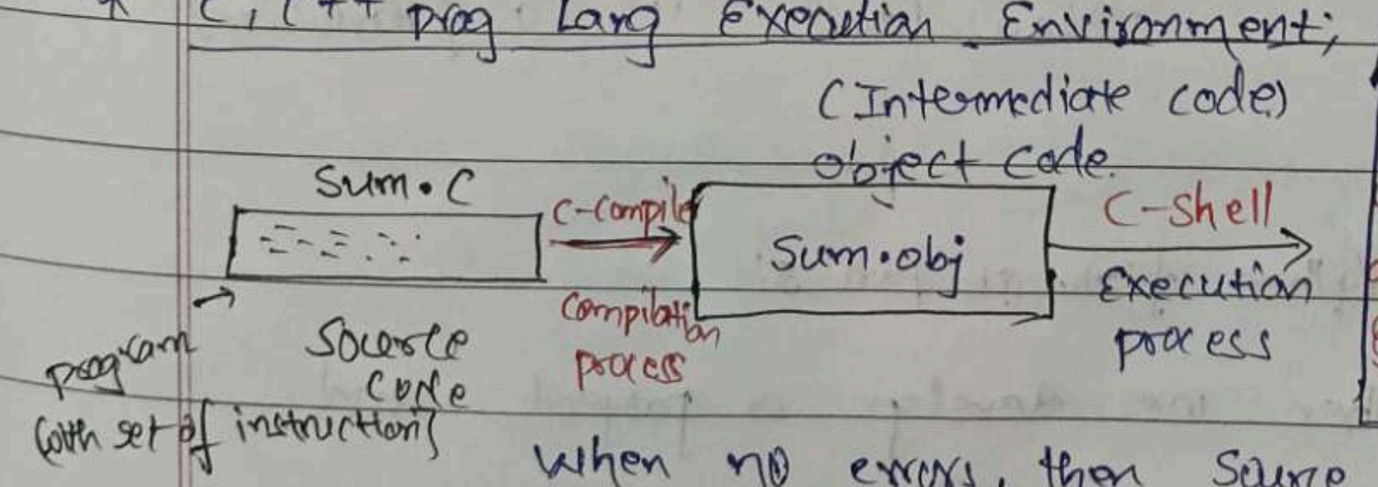


If the person converts line by line translation from Hindi to telugu then translator is nothing but Interpreter.

→ In next day newspaper; all meeting summary is present (translated) into a paragraph. This is done by Compiler.

→ Interpreter shows one error at a time but Compiler shows all the errors at a time.

* C, C++ prog Lang Execution Environment;

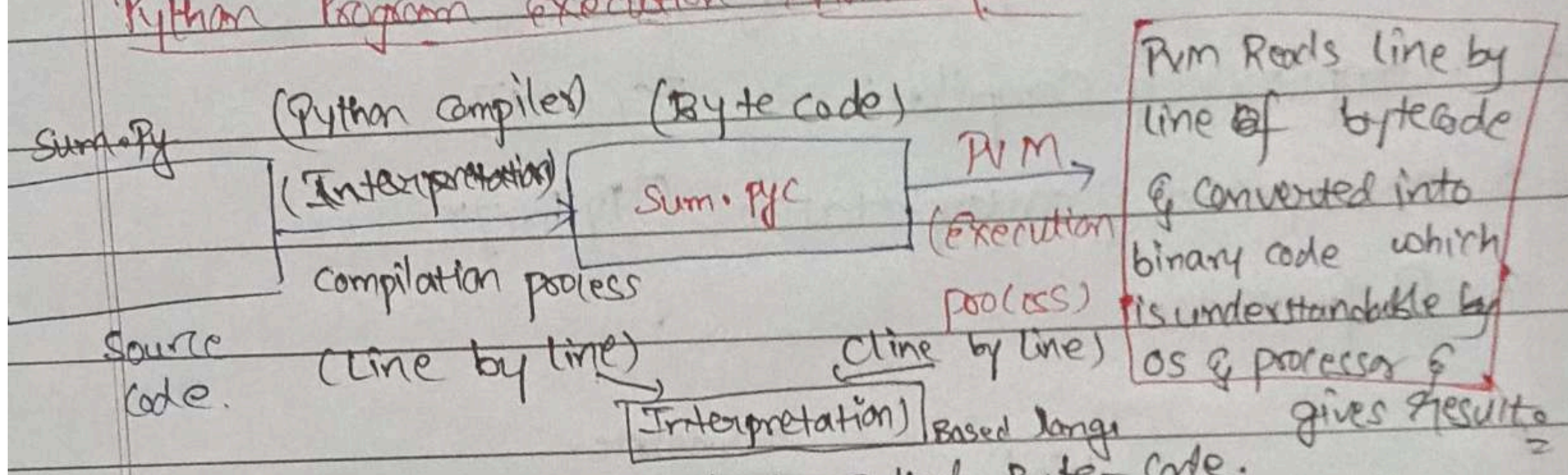


C-shell reads Entire object code & Converted into machine understandable code (binary code) and it is understandable by OS & processor & gives result

When no errors, then Source code will be converted into object code.

C-shell reads Entire object code & converted into machine code (binary)

Python Program execution Environment:



Python Intermediate Code is called Byte-Code.

PYC means Python Compiled Version.

If no errors, source code ⟶ Sum.pyC (Byte code)

→

Lec-9 - Python features - Part-4 [Day 3 - Go to Jupyter file]

5. Python is a High-level Programming Language.

6. Freeware & open Source:

Freeware = free to download

Human understandable, easy to write → High level language.

machine " → Low level " [OS & Hardware].

→

Lec10 - Python features - Part-5

Robust (strong): when we develop a project and after running it; the end user or Application user enters invalid i/p & by default we get Technical Errors messages which are understandable by Programmers but not by

end-users. So Programmer must develop/write the User-friendly error messages using Exception Handling.

* Python is Extensible & Embedded:

→ Extensible: Python provides its modules to other languages -- C, C++, html, Javascript etc.

* Embedded:

Python lang uses other language code segments/snippets.

*

lec-11- Variables in Python

→ Data Types: In Python, data means Any Value or information that a Program can store, process and manipulate."

Data Representation in Python:

Data: Processed Information (simply facts)

(or) studying the insights of information is called Data.

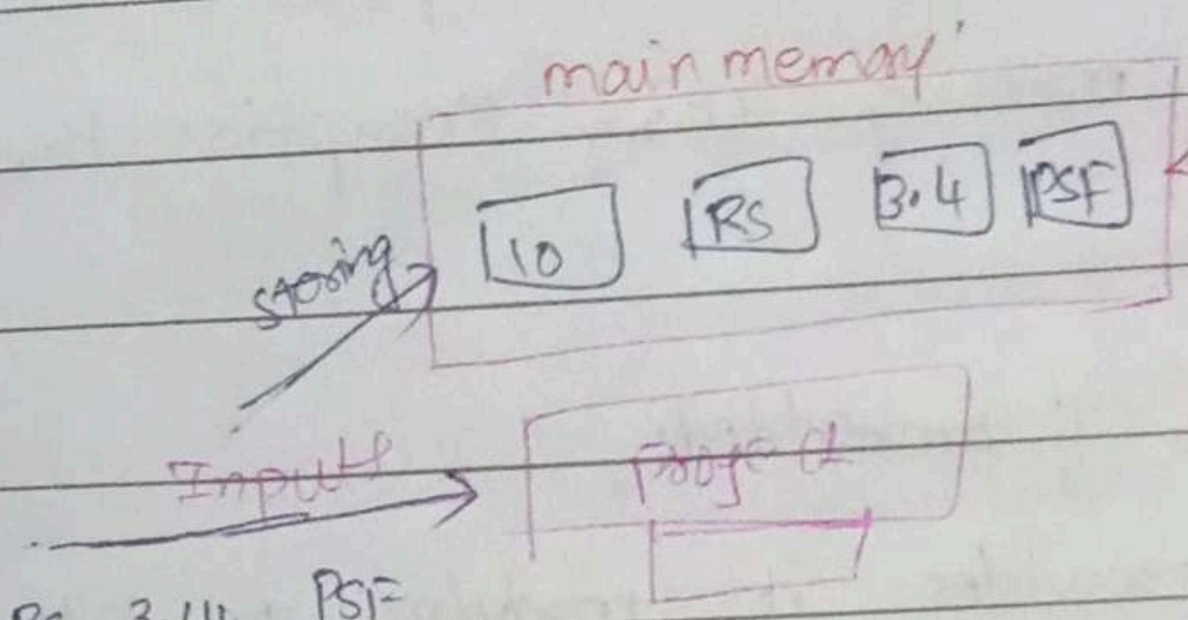
→ The purpose of data is to "analyze & take effective decisions in organization"

Ex: from this information (Text) I will gain some insights.

(Phanindra is the faculty of AI in MITS college)

Name	Faculty subject	college
Phanindra	AI	MITS

→ main memory (RAM)



10 Rs 3.14 PSF

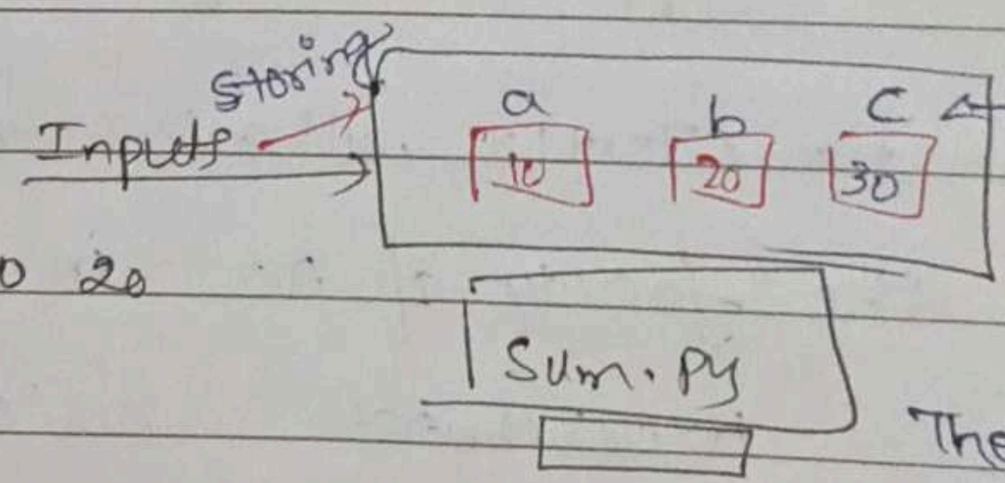
← By using data types memory space is allocated in main memory - so that data can be stored

* Literals & their Types:

→ In Python we have different types of literals or values or data

1. Integer literals
2. Float literals
3. String literal
4. Boolean literals
5. ~~Date literals~~
6. Collections literals.

Importance of identifiers (or) variables:



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← Here a, b, c are called Distinct Names given to the created memory spaces for identifying the values. These names are called identifiers

→ once values are stored, we need to process them. To process them, we need some reference here.

* Why identifiers are called Variables?

Here the values of identifiers are changing or varying during the program execution and hence identifiers are called identifiers.

$a=100$ previously $a=10$ & $b=20$ but now changed.

$b=200$

$c=a+b$

* In Python, all types of literals stored in the form of variables and all variables (values) in Python are called OBJECTS